VITAMIN B_6 (PYRIDOXINE)

Etiology and Pathogenesis

Vitamin B_6 refers to three interconvertible forms: **pyridoxine**, **pyridoxamine**, and **pyridoxal**. Humans cannot synthesize these compounds and must obtain them from the diet. Rich sources include meats, whole grains, vegetables, and nuts. Food processing can reduce the bioavailability of vitamin B_6, which is absorbed via passive diffusion in the jejunum. The primary active form is **pyridoxal-5-phosphate (PLP)**.

Vitamin B_6 serves as a coenzyme in numerous metabolic pathways, including: - Amino acid **decarboxylation** and **transamination** - **Gluconeogenesis** - Conversion of **tryptophan to niacin** - **Sphingolipid**, **prostaglandin**, and **neurotransmitter synthesis**

Because of its role in tryptophan-to-niacin conversion, clinical features of vitamin B_6 deficiency may resemble those of **niacin deficiency (pellagra)**.

Epidemiology

Dietary deficiency is rare due to widespread availability of vitamin B_6 in foods. However, deficiency can occur in: - **Alcoholics** (due to poor nutrition) - Patients with **malabsorption syndromes** (e.g., Crohn disease, celiac disease) - Individuals taking certain **medications**: - **Isoniazid**, **hydralazine**, **penicillamine**: bind to pyridoxal-5-phosphate, increasing excretion and reducing coenzyme activity - **Oral contraceptives**: may increase demand or reduce levels

Clinical Findings

Deficiency manifests with dermatologic, mucosal, neurologic, and systemic symptoms:

Dermatologic and Mucosal Features:

- **Seborrheic-like dermatitis** affecting the face, scalp, neck, shoulders, buttocks, and perineum
- **Photodermatitis**
- **Glossitis**: red, painful, ulcerated tongue with flattening of filiform papillae
- **Cheilitis** and **angular stomatitis (cheilosis)**
- **Conjunctivitis**

These findings constitute an **oculo-orogenital syndrome** similar to that seen in **riboflavin deficiency**.

Neurologic and Systemic Symptoms:

- **Peripheral neuropathy**, paresthesias, weakness
- **Confusion**, depression, irritability
- **Nausea, vomiting, anorexia**
- **Anemia** (due to impaired heme synthesis)

The clinical picture often mimics **pellagra** because vitamin B_6 is essential for converting tryptophan into niacin.

Toxicity:

Excessive intake of vitamin B_6 (usually from supplements) can cause **sensory peripheral neuropathy**, but typically does **not produce skin findings**.

Laboratory Testing

- **Plasma pyridoxal-5-phosphate (PLP)** is the standard test for assessing vitamin B_6 status.
- **Low PLP levels** confirm deficiency.

Treatment

- **Recommended daily intake**: - Adult males: 2 mg/day - Adult females: 1.6 mg/day - Infants: ~0.3 mg/day
- **Therapeutic dose**: 100 mg pyridoxine daily
- Discontinue offending medications if possible

Response to therapy: - Oral lesions resolve within **days** - Skin and hematologic abnormalities improve within **weeks** - Neurologic symptoms may take **months** to resolve

VITAMIN B_9 (FOLATE)

Etiology and Pathogenesis

Folate is abundant in many foods, especially: - Liver - Wheat bran and other grains - Leafy green vegetables - Dried beans

The active form, **tetrahydrofolate (THF)**, functions as a coenzyme in **single-carbon transfer reactions**, critical for: - Amino acid metabolism - Purine and pyrimidine synthesis (essential for DNA replication)

Causes of Deficiency:

- **Malabsorption**:
- Celiac disease
- Chronic diarrhea
- Post-gastrectomy state
- **Increased demand**:
- Pregnancy
- Hemolytic anemias
- Rapid cell turnover
- **Antifolate medications**:
- **Methotrexate**

- **Trimethoprim**
- **Pyrimethamine**
- **Oral contraceptives** (mild effect)

These agents interfere with folate metabolism or absorption, leading to functional deficiency.